

Release 26.4.22 (318103) — Code Review Report

Date: 2026-04-21

Release: KK PI 26.4.22 — [TP #318103](#)

Repos: MinuteMenu/KK, MinuteMenu/Centers-CX, MinuteMenu/POC_HxCloudConnectApi, MinuteMenu/Shared-Core, MinuteMenu/MinuteMenu.Database, MinuteMenu/HX, MinuteMenu/Parachute

Scope: 7 release-to-master PRs consolidating 45 TP tickets

Focus: Critical (90-100) and High (80-89) findings

Executive Summary

Severity	Count
Critical (90-100)	4
High (80-89)	17
Total	21

Missing tickets: 2 release-scope tickets have NO code change in any diff (#318581 Ghost eForm, #321662 Pending disallow). Flagged for product/engineering confirmation.

Critical Issues (Must Fix Before Release)

#	Repo	Area	TP Ticket	Category	Summary
C1	CX + KK	ACH re-download	#316306	Security/Bug	Missing authorization scope; SQL string-concat IN; multi-date batch stamped with single arbitrary date
C2	CX + KK	ACH re-download	#316306	FinancialAccuracy	Batch spanning multiple <code>check_date</code> values produces ACH file with only last row's date → bank compliance risk
C3	HX API + DB	J.017 policy	#320244	CrossRepoImpact	EF-mapped SponsorPolicy entity queries <code>include_infants_in_own_alone_edit_check_flag</code> ; any deploy before DB migration = HX API outage
C4	Parachute	Vehicle search	#320290	Bug	Unguarded cast <code>(double)e.PurchasePrice</code> on nullable decimal → 500 on search when any vehicle has null price

Critical Findings (Detail)

C1. **Centers-CX#4161 + KK#22699** — ACH re-download missing authorization + raw SQL IN

Ticket: [TP #316306](#) — Enable KidKare For Centers The Ability To Re-Generate Payments Without Needing To Void And Reissue

Score: 92 | **Category:** Security/Bug

File: `products/Centers/Projects/MinuteMenu.Centers.ServerLib/Logic/LogicServiceToKK.cs:65-113`

`( GetCenterACHFileFromCheckRegister ) +`

`Projects/KidKare.Service/Controllers/PortCX/PortCxPaymentController.cs:522-535 ( RedownloadACHFile )`

The ACH re-download SQL uses `WHERE check_id IN ( " + checkIdList + " )` by string-concatenating an `int[]` into raw SQL. While the int array prevents injection today, it breaks the parameterization pattern — any later change to the type erases that safety. More important, the controller performs no authorization check that the supplied `checkIds` belong to the current user's scope beyond `AND client_id = @client_id`. A sponsor user can force-download ACH batch files for arbitrary `check_id`s within the same client, and there is no null guard on `request.CheckIds` (`null` → 404 instead of 400).

Impact: Information disclosure (ACH bank account data for unrelated batches within same client); fragile non-parameterized IN list.

Recommendation: Parameterize the IN list (Dapper `@ids` or a TVP). Validate that all `checkIds` belong to the expected scope before building the ACH file. Cap `checkIds.Length` (e.g., ≤ 2000). Null-check `request` and `request.CheckIds`.

C2. **Centers-CX#4161** — ACH multi-date batch produces wrong effective date

Ticket: [TP #316306](#) — CX re-download ACH

Score: 90 | **Category:** FinancialAccuracy

File: `products/Centers/Projects/MinuteMenu.Centers.ServerLib/Logic/LogicServiceToKK.cs:71-95`

`( GetCenterACHFileFromCheckRegister )`

Inside the reader loop, `checkDate = Convert.ToDateTime(rdr["check_date"]).Date` is overwritten each row — the **last row's check_date wins** and is used for the whole ACH file. The KK batch-picker groups by `mod_date_time + mod_login_id` (not by `check_date`), so a single UI "batch" may legitimately contain rows with different `check_dates`. HX Homes' re-generate behavior preserves per-check posting dates; this fix does not. Additionally `cmd.CommandTimeout = 300` (5 min) is excessive for an indexed IN lookup.

Impact: Silent production of ACH files stamped with the wrong effective/settlement date. Since ACH files are submitted to banks, wrong-date files cause compliance/timing problems.

Recommendation: Take `checkDate` from the KK caller (`filter.checkDate`) via `RedownloadACHRequest` DTO; OR group rows by `check_date` inside the SP loop and emit one ACH file per distinct date; OR assert consistency across rows. Reduce `CommandTimeout` to 30s.

C3. [POC_HxCloudConnectApi#1595](#) + [MinuteMenu.Database#686](#) — EF entity read of DB column requires strict deploy order

Ticket: [TP #320244](#) — HX030 Error 76 disallow own infants

Score: 90 | **Category:** CrossRepolImpact

File: HxCloudConnect.Data/Entities/SponsorPolicy.cs (property added) + ≥14 call sites in ProviderService, AdministrationService, ChildService, ExportFileService, SponsorService

HX API SponsorPolicy EF entity gains `include_infants_in_own_alone_edit_check_flag` . EF generates `SELECT [all mapped columns] ON every db.SponsorPolicies.Where(...)` query. 14+ call sites hit this during routine traffic. If the HX API artifact deploys before DB migration `320244_ADD_NEW_FIELDS_TO_TABLE_SPONSOR_POLICY.sql` , every one of those queries throws `Invalid column name 'include_infants_in_own_alone_edit_check_flag'` → **full HX API outage**.

Companion risk: `c1sSPONSORPOLICY.c1s` in HX Desktop does `rsPopulate("include_infants_in_own_alone_edit_check_flag")` unconditionally. HX Desktop v3.4.0 installed on any sponsor whose DB has not yet been upgraded throws ADO "Item cannot be found" at sponsor-policy load — HX Desktop fails to open.

Impact: Any deploy-order violation = HX API outage (server side) and HX Desktop inoperable for affected sponsors (desktop side).

Recommendation: Enforce strict deployment order in the DevOps runbook: **DB Updates 320244_ADD + 320244_SET_DEFAULT must run BEFORE** any HX API artifact is promoted AND before HX Desktop v3.4.0 is pushed to any sponsor desktop. Add app-startup smoke test:

```
SELECT TOP 1 include_infants_in_own_alone_edit_check_flag FROM SPONSOR_POLICY . Consider wrapping the VB6 rsPopulate read in On Error Resume Next for tolerance.
```

C4. [Parachute#4330](#) — Vehicle search removes null-check on PurchasePrice

Ticket: [TP #320290](#) — Search Parity for Accounting Pages

Score: 90 | **Category:** Bug

File: Projects/Parachute.Service/Controllers/Expenses/MileageController.cs:113

The Vehicle search refactor removed the `e.PurchasePrice != null` guard. New line:

`isNumeric && SqlFunctions.StringConvert((double)e.PurchasePrice, 18, 2).Trim().Contains(stringNumber)` . Unconditional cast of nullable decimal → double throws `InvalidOperationException: Nullable object must have a value` during EF translation, or 500s the endpoint depending on provider behavior.

Impact: Any user searching the Large Inventory / Vehicles tab will see the entire endpoint 500 out if their account has **any** vehicle with a NULL purchase price. This is a very common data shape in production (legacy vehicles often lack purchase price).

Recommendation:

```
(e.PurchasePrice != null && isNumeric
    && SqlFunctions.StringConvert((double)e.PurchasePrice.Value, 18, 2)
    .Trim().Contains(stringNumber))
```

High Severity Findings (80-89)

Cross-Repo J.017 Policy

H1. MinuteMenu.Database#686 — save_provider_policy_sp silently overwrites highest_serving_allowed_code to 215

Ticket: TP #320881 — PCI Highest Serving Allowed empty

Score: 88 | **Category:** DataIntegrity

File: MMADMIN_XXX/Programmability/Stored Procedures/save_provider_policy_sp.sql and save_provider_policy_sp_port_kk.sql

Change from " 0 → NULL " to " 0 OR NULL → 215 " applies on **every update path**, not just INSERT. VB6 Integer parameters default to 0 when a form field was never loaded. Any existing HX Desktop update path that does not explicitly round-trip highest_serving_allowed_code now forcibly writes 215 over the stored value. HX API ProviderService.cs code-level fix is careful (loads existing value first); the SP change does NOT do that lookup.

Impact: Providers configured with Serving 2 or Serving 3 silently dropped to Serving 1 on next bulk edit → wrong meal allowance on next claim processing, cascades to reimbursement errors.

Recommendation: Only default to 215 in the INSERT branch of the SP:

```
IF @highest_serving_allowed_code = 0 OR @highest_serving_allowed_code IS NULL
    SELECT @highest_serving_allowed_code = ISNULL(highest_serving_allowed_code, 215)
FROM PROVIDER_POLICY WHERE provider_id = @id
```

Move the default to the app layer (as ProviderService.cs EnrollProvider already does).

H2. MinuteMenu.Database#686 — MRP manifest omits sponsor-030 enablement script

Ticket: TP #320244

Score: 80 | **Category:** CrossRepoImpact / DeploymentGap

File: MMADMIN_XXX/Updates/320244_SET_SPONSOR_030_INCLUDE_INFANTS_FLAG.sql + MRP.sql

The SET_SPONSOR_030_INCLUDE_INFANTS_FLAG.sql script (sets J.017='Y' for the ticket's requesting sponsor HX030) is committed to the repo but is **not listed in the regenerated MRP manifest**. MRP-driven deploy will not apply it → the feature ships disabled for the very client that asked for it.

Impact: Ticket appears complete but has no effect in production for HX030. Follow-up "not working" ticket expected.

Recommendation: Regenerate MRP.sql with the SET_SPONSOR_030 script included (sequenced after SET_DEFAULT so column is populated), OR document a manual DBA runbook step with product-owner sign-off.

Sponsor 102 Print Check + ACH (Cross-Repo)

H3. POC_HxCloudConnectApi#1595 + KK#22699 + Centers-CX#4161 + MinuteMenu.Database#686 — Cross-repo deploy-order contract

Ticket: TP #320213 — Sponsor 102 Print Check

Score: 85 | **Category:** CrossRepoImpact

File: CX LogicService.cs + CheckPaymentHistoryList.cs , KK PaymentMapping.cs + PaymentCheckRegisterResponse.cs , DB select_payment_history_list_sp.sql

select_payment_history_list_sp emits 3 new columns (mod_date_time, mod_login_id, mod_username). CX LogicService reads dr["mod_date_time"] . KK AutoMapper binds the new fields. KK consumes via the checked-in MinuteMenu.Centers.ServerLib.dll . Deploy ordering must be DB → CX service → KK → HX API; wrong order crashes the Check Register page for every sponsor with IndexOutOfRangeException .

Recommendation: Enforce deploy order in release runbook. Post-deploy smoke-test:

/cxservice/payments/getCheckRegisterList returns modDateTime/modLoginId/modUsername .

H4. POC_HxCloudConnectApi#1595 — Unbounded Task.WhenAll fan-out on Sponsor 102 print

Ticket: TP #320213

Score: 80 | **Category:** InfrastructureStress

File: HxCloudConnect.Api/Controllers/IssuePaymentController.cs + HxCloudConnect.Services/PaymentService.cs

Controller spins an unbounded Task.WhenAll calling BuildSponsor102Data **per check**. Each BuildSponsor102Data fires 5 stored procedures in parallel (active child, serving times, provider tier, child tier counts, training hours). A 500-provider "print all" = **2,500 concurrent DB round-trips** with no semaphore, no batching, no caching across providers that share fiscal/tier data.

Impact: Spike DB connections to hundreds; each QueryListByStoredProcedureAsync opens a new connection. Risk of SQL thread pool / connection pool exhaustion during monthly batch prints.

Recommendation: Add SemaphoreSlim(10) around sponsor102Tasks . Cache SELECT_PROVIDER_SERVING_TIMES_SP / tier data per providerId within the request. Monitor sp_who2 during first 102 print.

H5. POC_HxCloudConnectApi#1595 — EF-tracked entity mutation in AggregateLinkedClaimsForSponsor102

Ticket: TP #320213

Score: 80 | **Category:** DataIntegrity

File: HxCloudConnect.Services/PaymentService.cs:542-596

Fetches HxClaim rows for a check and mutates the chosen primary claim **in-place** by summing counts and amounts from siblings. The entity is EF-tracked — any subsequent SaveChanges() in the same request (or caching by IRepository) persists the summed values to DB, **corrupting the original claim row**.

Recommendation: `AsNoTracking()` on the initial query, or detach the entity, or clone to a new `HxClaim` for transient view logic.

H6. [POC_HxCloudConnectApi#1595](#) — `Sponsor102LinkedClaimDTO.ClaimYear` typed decimal?

Ticket: [TP #320213](#)

Score: 82 | **Category:** Bug

File: `HxCloudConnect.Data/DTO/Payment/PrintReportDTO.cs:359-367`

`ClaimYear` and `StatusCode` declared as `decimal?`. On the FE, `moment({ year: lc.ClaimYear, month: lc.ClaimMonth - 1, day: 1 })` and `getClaimLabel(lc.StatusCode, ...)` expect integers. Works today via JS coercion but semantically wrong — any future JSON serializer change (scientific form for large decimals) breaks the check stub.

Recommendation: Change `ClaimYear` to `int?` and `StatusCode` to `int?` to match `HxClaim.claim_year/claim_status_code`.

Enrollment & eForm Cleanup

H7. [KK#22699](#) — Cleanup endpoint fan-out with no `CommandTimeout`

Ticket: [TP #319630](#) — CX Child Not Found on eForm approval

Score: 88 | **Category:** InfrastructureStress

File: `Projects/KidKare.Bll/Enrollment/CxEnrollmentBll.cs:1444-1628`

`DetectOrphanedCxEforms` / `CleanupOrphanedCxEforms` execute a **serial per-client fan-out**. Each iteration opens a fresh `eFormsContext` + `CxContext`, then runs `GetChildQuery(cx, clientId, null, null).Select(x => x.child_id).ToList()` — pulls the **entire active CHILD roster** for the client into memory. No `CommandTimeout`. No row cap. Auto-detect branch (`null ClientIds`) sweeps every CX client with any active eForm assignment.

Impact: N-query sweep where N can be hundreds of clients. Connection pool pressure, long-running thread held for the duration, OOM risk on large tenants. Matches known incident patterns 3/4.

Recommendation: Require non-empty `clientIds` in prod (or cap per invocation). Add explicit `Database.CommandTimeout` of 30-60s. Move to a Hangfire/queue job rather than sync HTTP. Replace the `HashSet`-based orphan check with a server-side `NOT EXISTS` anti-join.

H8. [KK#22699](#) — "Orphan" definition includes withdrawn children → mass hard-delete risk

Ticket: [TP #319630](#)

Score: 85 | **Category:** DataIntegrity

File: `Projects/KidKare.Bll/Enrollment/CxEnrollmentBll.cs` (orphan predicate) + `EnrollmentBll.DeleteEformAssignments`

Orphan predicate: `a.ChildId == 0 || !activeChildIds.Contains(a.ChildId)`. `activeChildIds` is populated by `GetChildQuery` which filters `record_status_code == Active`. **Any child whose status is Withdrawn or Inactive is treated as an orphan** and its eForm assignment is HARD-deleted via `HardDelete()` on `EnrollmentForm`, history, signatures, IEF, reports, and storage files.

Impact: Production data loss. A child withdrawn yesterday (a common lifecycle event) will have all historical enrollment forms/IEFs permanently deleted on first cleanup run.

Recommendation: Tighten predicate to "no matching CHILD row at all" (drop the Active filter for this check). Explicitly exclude withdrawn/inactive children from cleanup scope. Require `confirm: true` token before delete. Consider soft-delete instead of hard-delete.

H9. KK#22699 — Admin cleanup endpoint authz is shared secret only

Ticket: TP #319630

Score: 82 | **Category:** Authorization

File: `Projects/KidKare.Service/Controllers/CommonController.cs:7436-7472`

`common/cleanupOrphanedEforms` and `common/detectOrphanedEforms` are guarded **only** by a shared-secret `SecurityKey`. No `[RequiredRole]`, `[RequiredPermission]`, `[Authorize]`, or IP restriction. If the key leaks (logs, Postman collections, client-side bundles), an attacker can hard-delete eForm data org-wide by omitting `clientIds` (auto-sweep mode).

Recommendation: Add `[RequiredRole(Role.MM.System.Admin)]` in addition to the security key. Move behind VPN / IP allow-list. Log every invocation with caller identity, `ClientIds`, and row counts.

Localization & Data Patches

H10. KK#22699 — Spanish warning `ACTIVE_CHILDREN_WARNING_SETIEF_SKIPPED` left in English

Ticket: TP #298391 — Spanish translations

Score: 85 | **Category:** Localization

File: `Projects/KidKare.Web/app/i18n/es/states/cx-active-children.json:28`

The key `ACTIVE_CHILDREN_WARNING_SETIEF_SKIPPED` was added to the Spanish file with the English text verbatim. Shown every time a sponsor performs Set IEF and at least one child lacks a current enrollment date.

Recommendation: Translate to Spanish.

H11. KK#22699 — Spanish `{{participant_kids_plural}}` placeholder dropped

Ticket: TP #298391

Score: 80 | **Category:** Localization

File: `Projects/KidKare.Web/app/i18n/es/states/my-kids-enroll-child.json:165`

Spanish translation hard-codes "varios miembros de una familia" instead of interpolating `{{participant_kids_plural}}` . Singular `{{participant_kids}}` kept — inconsistent. For sponsors configured with a non-default participant term ("Estudiantes", "Participantes", "Adultos"), the Spanish sentence loses terminology customization and also conveys a different meaning ("family members").

Recommendation: Restore the plural placeholder.

H12. MinuteMenu.Database#686 — Foster Income-Eligibility restore too narrow

Ticket: [TP #321413](#) — Restore Historical IE Dates for Foster Children

Score: 80 | **Category:** RequirementWrong / DataIntegrity

File: MMADMIN_XXX/Updates/321413_Restore_Foster_Child_Income_Eligibility.sql:72-101

WHERE clause narrows to `bk.tier_code = 126 AND bk.tier_start_date IS NOT NULL AND bk.tier_end_date IS NOT NULL` . Three gaps versus ticket scope:

1. Tier I only — foster children who legitimately had Tier II (127) or other `tier_code` data cleared by #318945 are **not** restored.
2. RCA reference uses `OR` between start/end dates; foster children with only one date populated are skipped.
3. `is_child_income` checkbox is NOT restored — ticket explicitly asks: "Restore checkbox state if applicable".

Recommendation: Broaden to

`bk.tier_code IS NOT NULL AND (bk.tier_start_date IS NOT NULL OR bk.tier_end_date IS NOT NULL)` . Also restore `is_child_income` from the backup table. Confirm scope with HX102 before release.

Parachute

H13. Parachute#4330 — MileageModel.Miles API contract break (decimal → double)

Ticket: [TP #320290](#)

Score: 85 | **Category:** API Contract

File: Projects/Parachute.Service/Models/MileageModel.cs:13

`Miles` property type changed from `decimal` to `double` . This alters the JSON serialization contract for `/mileage` . Strongly-typed mobile clients (iOS `Decimal` , Android `BigDecimal`) may fail or lose precision. Values now serialize with IEEE-754 artifacts like `12.340000000000001` .

Recommendation: Revert to `decimal` or add `MilesDouble` as a new field while keeping the existing contract. Perform internal conversion only where needed.

H14. Parachute#4330 — ExpensesController NullReferenceException on search

Ticket: [TP #320290](#)

Score: 80 | **Category:** Bug

File: Projects/Parachute.Service/Controllers/Expenses/ExpensesController.cs:509

New predicate `e.Category.Name.ToLower().Contains(req.InputSearch)` without a null guard. `ExpenseModel.Category` is a reference type and the same file elsewhere defensively accesses `category?.Name`.

Impact: 500 on search when any expense has a null `Category` (legacy data, deleted categories).

Recommendation:

```
(e.Category != null && !string.IsNullOrEmpty(e.Category.Name) && e.Category.Name.ToLower().Contains(req.InputSearch))
```

H15. [Parachute#4330](#) — number-input directive empty→null semantic change

Ticket: [TP #320293](#)

Score: 80 | **Category:** Bug

File: Projects/Parachute.Web/app/common/directives/number-input/number-input.js:86-92

`$parsers` now returns `null` for empty input when `allowNull=false` (previously fell through to `NaN` via `parseFloat('')`). Consumers calling `.toFixed()`, arithmetic, or `ng-pattern` on non-nullable models may throw. `$formatters` also changed — fields previously showing "0.00" now render blank.

Impact: Cross-cutting — affects kiosk PIN/hours entry, payment amount, payroll hours, expense amount, mileage, recurring invoice amounts. Forms that guard against `NaN` but not `null` will silently submit `null`; `ng-required` now treats `empty-valid`.

Recommendation: Return `undefined` (so required validation still marks invalid), OR keep `NaN` fallback when `!allowNull`. Regression test across kiosk PIN/hours, payroll, payment, expense amount, mileage.

H16. [Parachute#4330](#) — .search-input CSS class collides with existing <input> usage

Ticket: [TP #320291](#)

Score: 75 (just below High; flagging for visibility)

File: Projects/Parachute.Web/app/common/directives/search-input/search-input.less

The rewritten stylesheet redefines `.search-input` as a flex wrapper (`display: inline-flex; margin-right: 16px`). This is imported globally. Many existing pages apply `class="search-input"` directly to `<input>` elements (Messages sent view, Payer Invoice List, drag-and-drop lists, UI sandbox, expenses-income main-view). These inputs now receive flex/margin/position that was never intended.

Recommendation: Namespace: `app-search-input` `.search-input { ... }`, OR rename the wrapper class. (Listed here despite 75 score because the blast radius spans unchanged modules.)

Claims Engine

H17. MinuteMenu.Database#686 + HX#1676 — Default flip in select_rpt_meal_count_attendance_sp

Ticket: TP #318686 — Meal Counts & Attendance with FRP missing Breakfast/Lunch

Score: 78 (just below High — included for visibility)

File: CXADMIN/Programmability/Stored Procedures/select_rpt_meal_count_attendance_sp.sql:19

The in-SP default `@respectDisallowance` was flipped from `1` to `0`. Behavior is preserved only because `ISNULL(..., 'N')` fallback re-creates prior behavior when the `CLIENT_POLICY` row is absent. The pattern is fragile: any future refactor removing the `ISNULL` silently enables disallowed meals in the FRP report for every CX client.

Recommendation: Keep the variable default = `1` and only flip to `0` when the new L11 policy is explicitly `'Y'`. Pattern:
IF `@includeAllRecordedMeals = 'Y'` SET `@respectDisallowance = 0`; .

Infrastructure Impact Assessments

Batch 1 — Claims Engine & Meal Attendance

Risk Level: Medium

Summary: Changes are mostly additive but `claim_error_meal_attendance_count_fn` was fully rewritten (ticket asked only for Error 76 infant handling), and `select_rpt_meal_count_attendance_sp` flipped a default — both create latent regression/default-safety risks visible only at scale.

Details:

- `claim_error_meal_attendance_count_fn` Error 76 branch adds `PROVIDER+SPONSOR_POLICY` joins per call → scales with $\#(\text{claim errors}) \times \#(\text{reprocess runs})$
- `claim_error_meal_attendance_source_form_fn` adds `MEAL_ATTENDANCE` scan — single-scan, fine
- `select_361127_Child_Reconciliation_sp` adds a second full-scan `UPDATE` over `#PGRID` for `RECTYPE='A'` (duplicated across `MMADMIN_XXX` + `CXADMIN`)
- HX `clsClaimProcessor.cls` refactored for #321408; restored `iMiscGroupOverCount` in disallow formula
- Policy L.11 + policy seed default 'N' (legacy behavior preserved)
- `save_sponsor_policy_sp` parameter-add — safe default; deploy-order caveat applies

Tables affected: `CLAIM_ERROR`, `MEAL_ATTENDANCE`, `CHILD`, `PROVIDER`, `SPONSOR_POLICY`, `CLIENT_POLICY`, `POLICY`, `#PGRID`

Rollback difficulty: Medium (SPs rollback clean; HX client rollback is installer-level)

Monitoring: Watch execution time of `select_claim_error_summary_sp`, `select_claim_errors_for_rpt_sp`, `select_361127_Child_Reconciliation_sp`, `select_rpt_meal_count_attendance_sp`; HX crash logs for `IncludeInfantsInOwnAloneEditCheckFlag` binding errors

DevOps Action Required: Yes — deploy `MMADMIN_XXX` migrations + `save_sponsor_policy_sp` BEFORE HX installer; monitor post-deploy execution-time regression on Error 76 reports.

Batch 2 — J.017 "Include-Infants" Policy (Cross-Repo)

Risk Level: Critical

Summary: 4-repo schema change (DB, HX API, HX Desktop, KK). Deployment order is NOT self-protecting. Out-of-order deploy = outage or silent data corruption. `save_provider_policy_sp` can silently overwrite `highest_serving_allowed_code` to 215 across all existing providers on next VB6 save. MRP manifest missing the sponsor-030 enablement script — feature will not activate for HX030.

Tables affected: SPONSOR_POLICY, PROVIDER_POLICY, STATE_POLICY, STATE_POLICY_TD view; any EF query over SponsorPolicies in HX API.

Rollback difficulty: High — HX Desktop installer rollback is per-machine; column-add requires DROP COLUMN for full reversal.

DevOps Action Required: Yes — critical

1. **Mandatory deploy order:** MMADMIN DB → HX API → HX Desktop → KK. Block reverse order in release pipeline.
2. **Add** `320244_SET_SPONSOR_030_INCLUDE_INFANTS_FLAG.sql` **to MRP** before ship, OR document manual run with product-owner sign-off.
3. **#320881 regression test:** snapshot `PROVIDER_POLICY.highest_serving_allowed_code` distribution before release; simulate HX Desktop save-with-unchanged-form; diff should show zero changes. If Desktop overwrites to 215, block release.
4. **Post-deploy smoke:** HX Desktop login for HX030, HX102; KK sponsor policy page; Save Sponsor Policy round-trip; OER render on claim with >50 errors.
5. **Monitoring:** HX API for Invalid column name 'include_infants_in_own_alone_edit_check_flag' and VB6 ADO "Item cannot be found" in the first 30 min window.

Batch 3 — Sponsor 102 Print Check & Payments

Risk Level: High

Summary: Financial-path change across 4 repos with mandatory deploy order DB → CX → KK → HX API. Wide covering index on multi-million-row CHECK_REGISTER without `ONLINE=ON`. 100k-row browser fetch for dropdown. Unbounded fan-out of 5 SPs per provider on Sponsor 102 print. Cross-repo contract via compiled `MinuteMenu.Centers.ServerLib.dll` + SP signature.

Tables affected: CHECK_REGISTER (hot — index DDL window needed), `SELECT_ACTIVE_CHILD_`, `SELECT_PROVIDER_SERVING_TIMES_`, `SELECT_PROVIDER_TIER_`, `SELECT_CHILD_TIER_COUNTS_`, `SELECT_TRAINING_HOURS_*`.

Rollback difficulty: Medium-High — `MRP_Rollback.sql` only covers `select_payment_history_list_sp`; other SPs + L11 policy INSERT lack rollback.

DevOps Action Required: Yes

1. `sp_BlitzIndex 'dbo.CHECK_REGISTER'` in QA before deploy.
2. Add `WITH (ONLINE = ON, MAXDOP = 4)` to 316306 index creation if Enterprise Edition, else maintenance window.
3. Enforce DB → CX → KK → HX API deployment order in runbook.
4. Extend `MRP_Rollback.build.config` before GA OR document that rollback requires DBA intervention.
5. Stage a SQL connection / thread-pool monitor before first Sponsor 102 print post-release.
6. `sys.dm_db_index_usage_stats` check 7 days post-release — drop index if unused.
7. Smoke-test `/cxservice/payments/getCheckRegisterList` contains `modDateTime/modLoginId/modUsername`.

Batch 4 — Enrollment & eForm Cleanup

Risk Level: High

Summary: Two admin endpoints perform serial per-client sweep of eForms + CXADMIN with no timeout guard, unbounded row fetches, hard-deletes based on "orphan = non-Active child" semantics (a common lifecycle event), and shared-secret-only authz. Cross-cutting API shape change (`UpdateIEFChildrenByChildIds` `bool` → `object`) and JS notification signature change.

Rollback difficulty: Low for individual features; impossible for data hard-deleted via cleanup.

DevOps Action Required: Yes — critical gate

1. **BLOCK deployment of the cleanup endpoint** until the orphan predicate is tightened (or replaced with explicit orphan ID list).
2. Add explicit `Database.CommandTimeout` (30-60s) on both contexts in the per-client loop.
3. Add `[RequiredRole]` on top of `CheckSecurityKey` ; firewall the endpoint to DevOps app pool / IP allow-list.
4. Rotate API security key before/after first prod use. Audit-log every invocation.
5. Run initial sweep off-peak under SQL monitor.
6. Audit `advancedErrorNotification` callers for the title→conf signature change.

Batch 5 — Reports, UI, Localization

Risk Level: Medium

Summary: Mostly low-risk report swaps, Spanish i18n, UI patches. Introduces the two admin cleanup endpoints (same risk as Batch 4). New covering index on `CHECK_REGISTER` (same risk as Batch 3). Shared-Core `.rpt` changed without nuspec version bump (process gap, not runtime risk).

DevOps Action Required: Yes

1. Run `316306_create_check_register_client_center_covering_index.sql` DDL in maintenance window.
2. Firewall/IP-restrict the two new cleanup endpoints.
3. Verify `CHILD_TIER_BACKUP_318945` exists on each HX sponsor DB before running `321413_Restore_Foster_Child_Income_Eligibility.sql` ; capture before/after counts; run in explicit transaction.
4. Confirm Shared-Core reporting service website is redeployed (file replacement, not NuGet).
5. Monitor CX API thread count + CXADMIN/EForm connection count for 48h post-deploy.

Batch 6 — Parachute

Risk Level: Medium-High

Summary: UI refresh (standardized search + cell highlighting) + targeted accounting fixes. Highest risks: silent API contract change on `MileageModel.Miles` (decimal→double), NRE in expenses search, semantic change in shared `number-input` directive reaching beyond the ticket scope (kiosk/payments/payroll), `.search-input` CSS class collision with existing inline usage.

DevOps Action Required: Yes

- 1. **Pre-deploy:** verify mobile clients (iOS/Android) handle double Miles without precision/crash; gate behind contract version if needed.
- 2. **Pre-deploy:** apply null guard on e.Category.Name in ExpensesController — prevents guaranteed 500 on null-category legacy data.
- 3. **Pre-deploy smoke:** number-input clear/empty behavior in kiosk PIN/hours, payment amount, mileage, expense amount.
- 4. **Pre-deploy visual regression:** pages using <input class="search-input"> directly (Messages, Payer Invoice List, drag-and-drop, UI sandbox, expenses).
- 5. **Post-deploy APM:** NullReferenceException on ExpensesController.GetExpense ; JSON deserialization errors on /mileage .
- 6. **Perf watch:** RUM/LCP on large expense/income tables.
- 7. **Follow-up:** remove legacy common/directives/search-input/* + index.html script tag.

Missing Tickets — No Code Change Found

Two tickets are in the release scope but no matching diff content was found in any of the 7 repos reviewed:

Ticket	Title	Status	Concern
TP #318581	Ghost eForm Enrollment Record (Not Started)	Dev	No enrollment-status counting / View Status change in any diff. #319630 orphan framework operates on EnrollmentAssignmentsCx only — HX assignments (this ticket is for HX provider 019603522) are not scanned.
TP #321662	Unable to disallow Pending children after Withdrawal	Ready For Test	Searched all diffs for "pending/withdraw/disallow/D.19/F.5b" — no matching change. HX claim processor changes are for #321408 and #320244 only.

Recommendation: Confirm with engineering whether these are in scope. If yes, add the missing code. If not, remove from the release entity so the release scope is accurate.

Priority Action Items

Must Fix (Critical — blocks release)

- 1. **C1** — [Centers-CX#4161](#) ([TP #316306](#)): Parameterize the ACH check_id IN (...) , add scope authorization on CheckIds , null-check the request.
- 2. **C2** — [Centers-CX#4161](#) ([TP #316306](#)): Validate all batch rows share the same check_date (or group by date server-side); reduce CommandTimeout to 30s.
- 3. **C3** — [POC_HxCloudConnectApi#1595](#) + [MinuteMenu.Database#686](#) ([TP #320244](#)): Enforce deploy order (DB → HX API → HX Desktop) in release runbook. Add app-startup smoke test for the new column.
- 4. **C4** — [Parachute#4330](#) ([TP #320290](#)): Restore null guard on e.PurchasePrice in MileageController.cs:113 .

Should Fix (High — significant risk)

1. **H1** — [MinuteMenu.Database#686 \(TP #320881\)](#): Scope `save_provider_policy_sp` 215 default to INSERT-only; preserve existing DB value on UPDATE.
2. **H2** — [MinuteMenu.Database#686 \(TP #320244\)](#): Add `320244_SET_SPONSOR_030_INCLUDE_INFANTS_FLAG.sql` to MRP manifest.
3. **H3** — Cross-repo ([TP #320213](#)): Document DB → CX → KK → HX API deploy order. Post-deploy smoke `/cxservice/payments/getCheckRegisterList`.
4. **H4** — [POC_HxCloudConnectApi#1595 \(TP #320213\)](#): Add `SemaphoreSlim(10)` and caching to `BuildSponsor102Data` fan-out.
5. **H5** — [POC_HxCloudConnectApi#1595 \(TP #320213\)](#): `AsNoTracking()` or detach in `AggregateLinkedClaimsForSponsor102`.
6. **H6** — [POC_HxCloudConnectApi#1595 \(TP #320213\)](#): Change `ClaimYear` and `StatusCode` to `int?`.
7. **H7** — [KK#22699 \(TP #319630\)](#): Require `clientIds`, add `CommandTimeout`, move cleanup to a Hangfire job.
8. **H8** — [KK#22699 \(TP #319630\)](#): Tighten orphan predicate to "no matching CHILD row at all". Require explicit confirm token.
9. **H9** — [KK#22699 \(TP #319630\)](#): Add `[RequiredRole]` on top of `CheckSecurityKey`; firewall behind VPN.
10. **H10** — [KK#22699 \(TP #298391\)](#): Translate `ACTIVE_CHILDREN_WARNING_SETIEF_SKIPPED` to Spanish.
11. **H11** — [KK#22699 \(TP #298391\)](#): Restore `{{participant_kids_plural}}` placeholder in Spanish.
12. **H12** — [MinuteMenu.Database#686 \(TP #321413\)](#): Broaden `321413_Restore_Foster...sql` WHERE (drop Tier-I filter; OR instead of AND). Restore `is_child_income`.
13. **H13** — [Parachute#4330 \(TP #320290\)](#): Revert `MileageModel.Miles` to decimal OR add `MilesDouble` as additive.
14. **H14** — [Parachute#4330 \(TP #320290\)](#): Null-guard `e.Category.Name` in `ExpensesController.cs:509`.
15. **H15** — [Parachute#4330 \(TP #320293\)](#): Review `number-input` empty→null change; either return `undefined` or keep `NaN`; regression-test all callers.

Verify With Product (Behavioral changes needing confirmation)

1. **#318686** — Policy L.11 defaults to 'N' (legacy broken behavior). Sponsor must opt-in per client. Is this intentional or should the fix be auto-applied to the affected client?
2. **#318581 & #321662** — No code change found in any diff. Confirm whether these tickets are meant for 26.4.22 or should be moved out of scope.
3. **#318784** — Fix is a workaround (frontend skips `licensed` auto-override on roster reports) rather than reading historical At-Risk data. Confirm product intent.
4. **#319630** — Delivered as general-purpose orphan framework scanning all CX clients. Original ticket was targeted to 2 children on client 8141. Confirm broader scope was intended.
5. **#320290 income date format** — Changed from MM/DD/YY to MM/DD/YYYY on the Income grid. Intentional?
6. **#320293 deleteFileReceipt call-site change** — Unrelated to any of the 5 Parachute tickets. Confirm QA tested delete-expense with attached receipt.
7. **#321408 isSchoolAgedDisallowCount storage under non-disallow waivers** — New gating may zero out a count that was previously written under `bAttendanceFromTimes=True` + Warn/Ignore waiver. Verify with HX211 scenarios.
8. **#321350 full-function rewrite of claim_error_meal_attendance_count_fn** — Non-76 error branches restructured; run side-by-side regression on OER output pre/post for production-scale claim-errors.

Report generated by 18 parallel code-review agents (6 batches × 3 specialized agents: Bug Detection, Architecture & Performance, Requirement Alignment) reviewing 7 release-to-master PRs. Only Critical (90-100) and High (80-89) findings are shown above; approximately 25 additional findings scored 70-79 (Notable) were filtered out — ask to include them if desired.